

WEEK 4



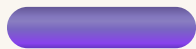
Nested Loops

A loop inside a loop.

중첩 반복문 — 반복문 속 반복문 · 영어코딩

BIG PICTURE

From a row to a whole floor



One loop = a line. A loop inside a loop = a square.

줄 하나 → 바닥 전체

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

Outer + inner

```
for row in range(3):      # outer: rows
    for col in range(4):  # inner: a row of blocks
        agent.place(DOWN)
        agent.move(FORWARD, 1)
    agent.turn(RIGHT)      # start the next row
```

Inner finishes fully before the outer steps once.

CONCEPT

How many times total?



outer × inner

$$3 \times 4 = 12$$



blocks placed

12 blocks

The inner loop runs **every time** the outer loop steps.

WATCH IT!

Inner loop fills each row

row 0

row 1

row 2

outer = 0

```
▸ each row = inner loop of 4
```

CODE

A square floor with a variable

```
size = 5

for row in range(size):
    for col in range(size):
        agent.place(DOWN)
        agent.move(FORWARD, 1)
    agent.turn(RIGHT)
```

5 × 5 = a 25-block square floor.

PREDICT 🤖

What shape, how many blocks?

Predict first — then click reveal

```
for row in range(4):  
    for col in range(4):  
        agent.place(DOWN)  
        agent.move(FORWARD, 1)  
    agent.turn(RIGHT)
```

Reveal answer 🔍 reveal

VOCAB

New words today

nested loop a loop placed inside another loop

outer loop the loop on the outside (rows)

inner loop the loop inside (one row of blocks)

grid rows and columns of blocks

중첩 반복문

바깥 반복문

안쪽 반복문

격자



DEBUG

What's wrong here?

buggy

```
# goal: a 5x5 square floor  
for row in range(5):  
    agent.place(DOWN)  
    agent.move(FORWARD, 1)  
    agent.turn(RIGHT)
```

Think first — then click next.



The fix

Bug: There's only *one* loop, so each "row" is a single block. A square needs an **inner loop** to fill each row before turning.

fixed

```
for row in range(5):  
    for col in range(5):  
        agent.place(DOWN)  
        agent.move(FORWARD, 1)  
    agent.turn(RIGHT)
```

YOUR TURN!

Build a 7×7 floor



Set `size = 7`



Outer loop for rows, inner loop for blocks



Turn right after each row



Stand on the floor and check the center



Send a photo + walkthrough on KakaoTalk



QUIZ

QUESTION 1 OF 3

In a 3×4 nested loop, how many blocks are placed?

12 (3×4).

3.

4.

7 ($3 + 4$).

 QUIZ

QUESTION 2 OF 3

What's missing to make a square floor?

```
for row in range(5):  
    agent.place(DOWN)  
    ...
```

More turn commands.

A variable named size.

An inner loop to fill each row.

A bigger range number.



QUIZ

QUESTION 3 OF 3

What does the **inner** loop do?

Counts the rows.

Builds one full row of blocks.

Turns the agent right.

Moves the agent up.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Nested loops mastered!

Next: stack the squares into your first pyramid.

다음 주: 사각형을 쌓아 첫 피라미드 완성